

Moses C. Nah

ROBOTICS, CONTROL THEORY, AND MOTOR NEUROSCIENCE

✉ moses890@gmail.com | 🏠 mosesandlily.github.io | 📷 MosesAndLily

Summary

I am a Postdoctoral Research Scientist at Honda Research Institute USA, developing self-improving control algorithms for contact-rich manipulation. I earned my Ph.D. at MIT under [Prof. Neville Hogan](#) and worked closely with [Prof. Jean-Jacques Slotine \[T01\] \[ArX02\]](#), with research recognized by Best Paper Awards at IROS 2024 [\[C01\] \[HA01\]](#) and BioRob 2020 [\[C04\] \[HA04\]](#). I also co-developed [Explicit™ \[C02\]](#), an open-source library leveraging Differential Geometry for rigid-body kinematics and dynamics, with [Dr. Johannes Lachner](#). Prior to MIT, I graduated *Summa Cum Laude* from Seoul National University with a B.S. in Mechanical and Aerospace Engineering. I am also a Korean Physics Olympiad Gold Medalist [\[HA09\]](#) and a selected candidate for the International Physics Olympiad national team.

Selected Professional Experience

Honda Research Institute USA

San Jose, CA, USA

POSTDOCTORAL RESEARCH SCIENTIST: ROBOTICS AND CONTROL SYSTEMS

Aug. 2025 – Present

- Division: Intelligent Robotics Research (IRR)
- Developing intelligent, self-improving control algorithms for contact-rich manipulation tasks

NAVER LABS Robotics Team

Gyeonggi-do, S.Korea

ROBOTICS RESEARCH INTERN

Aug. 2016 – Feb. 2017

- Supervisor: Dr. Sang-ok Seok
- Project 1: Development of a Wheel-Based Stair-Climbing Robot [\[C07\]](#)
- Project 2: Development of a Li-Ion Battery Pack PCB for Universal Use in NAVER LABS Robots [\[C06\]](#)

Selected Publications

JOURNAL

[J01] Johannes Lachner, **Moses C. Nah**, Neville Hogan. *A Physically Consistent Stiffness Formulation for Contact-Rich Manipulation*. The International Journal of Robotics Research (IJRR). 2025

[J02] **Moses C. Nah**, Johannes Lachner, Neville Hogan. *Robot Control Based on Motor Primitives: A Comparison of Two Approaches*. The International Journal of Robotics Research (IJRR). 2024

CONFERENCE

[C01] **Moses C. Nah**, Johannes Lachner, Federico Tessari, Neville Hogan. *On the Modularity of Elementary Dynamic Actions*. International Conference on Intelligent Robots and Systems (IROS). 2024. **Best Conference Paper Award** [\[HA01\]](#)

[C02] Johannes Lachner*, **Moses C. Nah***, Stefano Stramigioli, Neville Hogan. *Exp[licit]: An Educational Robot Modeling Software Based on Exponential Maps*. International Conference on Advanced Intelligent Mechatronics (AIM). 2024. ***Equal Contribution**.

[C03] **Moses C. Nah**, Aleksei Krotov, Marta Russo, Dagmar Sternad, Neville Hogan. *Manipulating a Whip in 3D via Dynamic Primitives*. International Conference on Intelligent Robots and Systems (IROS). 2021

[C04] Xiaofeng Xiong, **Moses C. Nah**, Aleksei Krotov, Dagmar Sternad. *Online Impedance Adaptation Facilitates Manipulating a Whip*. International Conference on Intelligent Robots and Systems (IROS). 2021

[C05] **Moses C. Nah**, Aleksei Krotov, Marta Russo, Dagmar Sternad, Neville Hogan. *Dynamic Primitives Facilitate Manipulating a Whip*. International Conference on Biomedical Robotics and Biomechatronics (BIOROB). 2020. **Best Student Paper Award** [\[HA04\]](#)

[C06] Dongil Choi, Minsu Kim, Hyeongkeun Kim, Choe Jonghun, **Moses C. Nah**. *Motion Planning of Autonomous Personal Transporter Using Model Predictive Control for Minimizing Non-Minimum Phase Behavior*. International Conference on Ubiquitous Robots (UR). 2018

[C07] Jonghun Choe*, Ukjin Kwon*, **Moses C. Nah***, Hyeongkeun Kim*. *Design Analysis of Tuskbot: Universal Stair-Climbing 4-Wheel Indoor Robot*. International Conference on Intelligent Robots and Systems (IROS). 2017. ***Equal Contribution**.

ARXIV

[ArX01] **Moses C. Nah**, Johannes Lachner, Neville Hogan. *Modular Robot Control with Motor Primitives*. 2025

[ArX02] **Moses C. Nah**, Johannes Lachner, Neville Hogan, Jean-Jacques Slotine. *Combining Movement Primitives with Contraction Theory*. 2025

[ArX03] Johannes Lachner, Federico Tessari, A. Michael West Jr., **Moses C. Nah**, Neville Hogan. *Divide et Impera: Decoding Impedance Strategies for Robotic Peg-in-Hole Assembly*. 2024

Education

Massachusetts Institute of Technology (MIT)

PH.D. IN MECHANICAL ENGINEERING

- Advisor: Prof. Neville Hogan
- Thesis: *Modular Robot Control with Motor Primitives* (Defense: Oct. 29, 2024)
- Committee: Prof. Jean-Jacques Slotine, Prof. Alberto Rodriguez
- Major: System Dynamics and Control
- Minor: Information Theory and Machine Learning

Cambridge, MA, USA

May 2020 – Dec. 2024

Massachusetts Institute of Technology (MIT)

M.S. IN MECHANICAL ENGINEERING

- Advisor: Prof. Neville Hogan
- Thesis: *Dynamic Primitives Facilitate Manipulating a Whip*
- Qualifying Exam Subjects: Dynamics, Control, Stochastic Systems

Cambridge, MA, USA

Sept. 2018 – May 2020

Seoul National University (SNU)

B.S. IN MECHANICAL AND AEROSPACE ENGINEERING

- Summa Cum Laude
- Leave of Absence for Military Service (2 years)

Seoul, South Korea

Mar. 2011 – Sept. 2018

Selected Honors & Awards

- 2024 [HA01] **Best Conference Paper Award**, IEEE/RSJ IROS (\$2,000)
- 2024 [HA02] **Sontheimer Travel Award**, Massachusetts Institute of Technology (\$1,500)
- 2023 [HA03] **MathWorks Fellowship**, Massachusetts Institute of Technology
- 2020 [HA04] **Best Student Paper Award**, IEEE BIOROB
- 2017 [HA05] **Young Talent Support Fellowship**, NAVER LABS
- 2017 [HA06] **Gwanak Fellowship**, Hanil Corporation
- 2013 [HA07] **Grand Prize**, SNU Design, Manufacturing Process, and Laboratory Contest
- 2011 [HA08] **First Runner-Up**, SNU Creative Engineering and Design Contest
- 2009 [HA09] **Gold Medalist**, 12th Korea Physics Olympiad (KPhO), High School Division

Teaching Experience

Teaching Assistant (MIT)

- Spring 2025 [T01] **2.152 Nonlinear Control**, Prof. Jean-Jacques Slotine (Rating: 7.0/7.0, Median)
- Fall 2024 [T02] **2.151 Advanced System Dynamics and Control**, Prof. Neville Hogan (Rating: 6.9/7.0)
- Fall 2022 [T03] **2.032 Dynamics**, Prof. Triantaphyllos Akyas (Rating: 6.8/7.0)
- Fall 2021 [T04] **2.151 Advanced System Dynamics and Control**, Prof. Neville Hogan (Rating: 6.8/7.0)

Presentations and Invited Talks

- 2025 [P01] **Harvard NSF Workshop on Reinforcement Learning**
- 2025 [P02] **North Carolina State University**, Topic: Modular Robot Control with Motor Primitives
- 2023 [P03] **IROS**, Workshop on Leveraging Models for Contact-Rich Manipulation (*OpenReview*)
- 2022 [P04] **KUKA Robotics**, Topic: Robot Control Based on Motor Primitives (Virtual)
- 2021 [P05] **Robotics: Science and Systems (RSS)**, Workshop on Deformable Object Simulation (DO-Sim)
- 2020 [P06] **Neural Control of Movement (NCM)**, Blitz Talk (*Program*)
- 2019 [P07] **MIT Embodied Intelligence Research Mixer**
- 2017 [P08] **TEDxSNU**, Featured Speaker, Session: *People Who Make Something*

Skills

Robotics & Mechatronics	KUKA LBR iiwa, Franka Emika Panda, Allegro Hand, ROS 1/2, LabVIEW, KiCAD, SolidWorks
Simulation & Modeling	MuJoCo, MATLAB
Programming & Tools	C/C++, Java, Python, Docker, Bash, Git, $\LaTeX$
Mathematics (Selected Topics)	Nonlinear Control Theory, Differential Geometry, Rigid Body Kinematics and Dynamics
Languages	Korean (Native), English (Full Professional Proficiency), Japanese (Elementary)